Integration of AI in sports management: navigating challenges and exploring opportunities

دمج الذكاء الاصطناعي في التسيير الرياضي: مواجهة التحديات واستكشاف الفرص

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Abstract:

The integration of AI into sports management represents a significant advancement, offering opportunities to optimize athlete performance, improve fan engagement, and streamline organizational operations. However, this transition comes with notable challenges, such as the need for adequate technological infrastructure, ethical concerns related to data collection, and resistance to change within management teams.

This study explores the various applications of AI in sports, including predictive analytics for player performance, personalization of spectator experiences, and automation of administrative processes. By navigating these challenges and capitalizing on opportunities, sports organizations can not only improve their efficiency but also redefine the future of sports management.

Keywords: AI; sports management; challenges; opportunities.
JEL Classification Codes: L83.

ملخص:

يتيح دمج الذكاء الاصطناعي في إدارة الرياضة فرصا لتحسين أداء الرياضيين، وزيادة تفاعل الجماهير، وتبسيط العمليات التنظيمية. ومع ذلك، يصاحب هذا التحول تحديات ملحوظة، مثل الحاجة إلى بنية تحتية تكنولوجية مناسبة، والمخاوف الأخلاقية المتعلقة بجمع البيانات، ومقاومة التغيير داخل فرق الإدارة.

تستكشف هذه الدراسة التطبيقات المختلفة للذكاء الاصطناعي في التسيير الرياضي، بما في ذلك التحليلات التنبؤية لأداء اللاعبين، وتخصيص تجارب المشاهدين، وأتمتة العمليات الإدارية، وقد توصلت الدراسة إلى أنه من خلال التغلب على التحديات واغتنام الفرص، يمكن للمنظمات الرياضية تحسين كفاءتها وإعادة تعريف مستقبل إدارة الرياضة.

كلمات مفتاحية: الذكاء الاصطناعي؛ التسيير الرياضي؛ التحديات؛ الفرص.
تصنيفات JEL : L83.

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1. INTRODUCTION

Historically, sport has been a field where human physical and mental abilities are constantly put to the test. From the ancient Olympic Games, which celebrated strength, speed, and endurance, to modern competitions, where strategy and innovation play a central role, sport has always reflected societal developments. Over time, technological advances have helped redefine the limits of human performance. The integration of technology and data analytics into sports management is becoming a fundamental pillar for strategic and operational decision-making. Coaches, sporting directors, and professionals now have access to a vast amount of information to make more informed decisions, improve player performance, and optimize team management. Today, AI goes far beyond these initial innovations. Its applications include predictive analytics to optimize training, early injury detection, and improving the spectator experience through advanced real-time analytics.

In the fast-paced world of professional sports, AI has become more than just a trend. The use of AI and data analytics has become a cornerstone for teams looking to gain an edge. The marriage of big data and machine learning has revolutionized the way teams recruit players, optimize game strategies, and even anticipate injuries. AI is no longer just a tool for analyzing statistics; it's shaping the very decisions that define the outcome of the game

1.1 Problem statement:

AI is transforming many industries, and sports is no exception. By analyzing vast amounts of data in real time, it is revolutionizing training strategies, optimizing athlete performance, and improving the spectator experience. Whether it's preventing injuries, adjusting game tactics, or automating sporting event management, AI is emerging as an essential strategic tool. Thanks to its learning capabilities, it can adapt decisions based on past performance and anticipate future needs, paving the way for a new era for the sports industry. However, the integration of AI in sport is not without controversy. Several ethical and practical questions arise.

The arrival of AI, with its advanced analytical capabilities and precise recommendations, raises a fundamental question: "**how far should AI be integrated into sports management? What are the opportunities for its use and the challenges it may face?".**

1.2 Importance of the study:

The study on the integration of AI in sports management is crucial because it allows for an analysis of the challenges and opportunities associated with the use of advanced technologies in this dynamic field. By examining how AI can optimize decision-making, improve the fan experience, and enhance athlete performance, this research contributes to a better understanding of the economic and ethical implications of AI.

1.3 Objectives of the study:

This study explores the integration of AI in sports management, its current applications, and the ethical challenges it raises. The objective is to determine whether the integration of AI represents a step forward toward a promising future or a distortion of the fundamental values that define sport.

1.4 Study Methodology:

Our research objective and our chosen approach led us to employ two methods:

- The descriptive method, based on the consultation of various books and theses, aimed at constructing a theoretical framework that will enable us to understand the basic concepts underlying our work.
- The analytical method was used to highlight the opportunities for integrating AI in sports management and to identify the most significant challenges that may be faced.

1.5 Research structure:

In order to answer the question asked and also to achieve the objectives of this study, we have structured our work around four (04) axes:

- About AI;
- Sports Management: Concept and Specificities;
- Opportunities offered by AI in sports management;
- current challenges of AI in sports management.

2. About AI

The basic idea of artificial intelligence is to understand context and make intelligent decisions based on available information.

AI therefore refers to the simulation of human intelligence in machines programmed to think and learn like humans. It encompasses various subfields, including machine learning, natural language processing, and computer vision. Machine learning is a key component of AI, where algorithms are designed to analyze and interpret data, allowing machines to make predictions or decisions without explicit programming. Natural language processing aims to enable machines to understand and interpret human language, while computer vision involves teaching machines to interpret and understand visual information (Djellaba & Benamara, 2023).

2.1 Definition

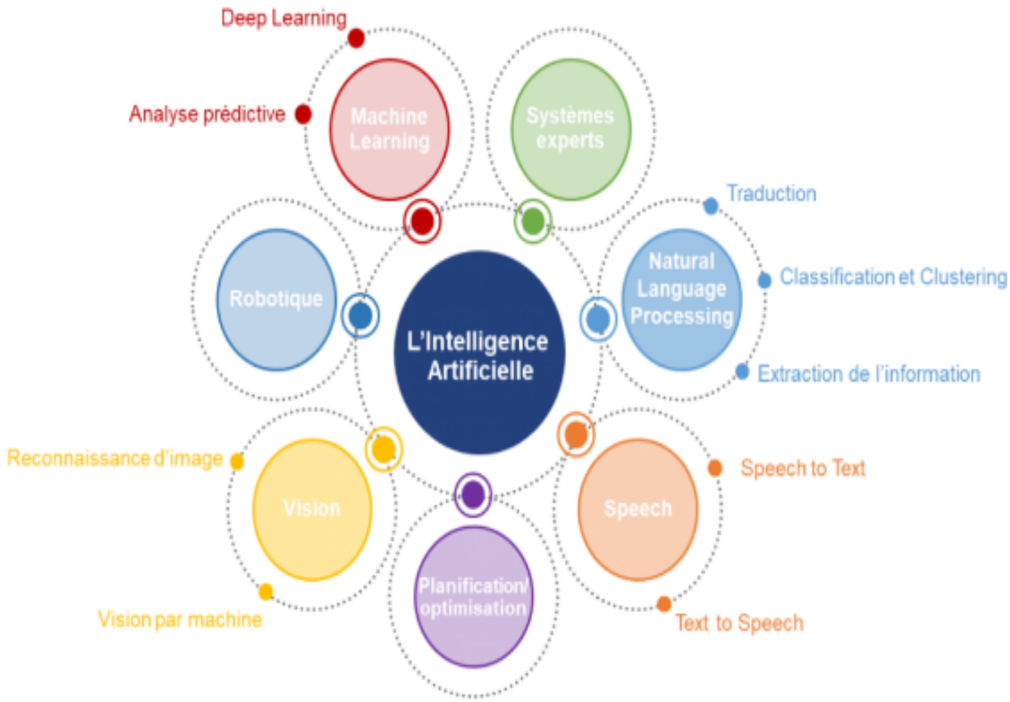
Several definitions of artificial intelligence have been proposed, some of which can be listed as follows:

- Kokina and Davenport (2017) consider AI as synonymous with cognitive technology or cognitive computing with the level of intelligence adapted to perform cognitive tasks (Kokina & Davenport, 2017, p. 118).
- Haenlein and Kaplan defined AI as the ability of a system to accurately understand external data, learn from it, and apply what it has learned to achieve specific goals and tasks through flexible adaptation (Rizvan Hasan, 2022, p. 443).
- Artificial intelligence (AI) includes the ability to reason, learn,

and adapt and can be widely used in business operations to automate tasks, decision-making, and customer relationship management (Hu, Chen, Hsu, & Tzeng, 2023, p. 2).

AI encompasses various subfields such as machine learning, natural language processing, and computer vision. These concepts enable machines to think and learn like humans, making predictions, interpreting language, and understanding visual information. AI has the potential to revolutionize industries and improve our daily lives, and its development continues to advance at a rapid pace (Djellaba & Benamara, 2023, p. 3). The following figure provides an overview of the most important areas of AI:

Fig.1. Overview of AI fields



Source: (Barraud, 2020, p. 30)

2.2 Applications

The combination of applications and technological fields gives us specific technologies such as (Rikhardsson, Thórisson, Bergthorsson, & Batt, 2022, p. 325):

- ✓ Inductive programming language that uses formal logic to represent a database of facts and formulate hypotheses derived from this data;

- ✓ Robotic Process Automation (RPA), which extracts a list of rules and actions to perform by observing the user perform certain tasks in the application's user interface in order to achieve certain goals;
- ✓ Expert systems that have coded rules to mimic the human decision-making process for reasoning and solving certain classes of problems;
- ✓ Decision networks comprise a set of variables and their probabilistic relationships that can process missing information to solve problems and improve decisions;
- ✓ Artificial neural networks that can learn to improve performance without receiving explicit instructions on how to do so, with the aim of improving decision-making, planning, and goal achievement;
- ✓ Autonomous systems are at the intersection of robotics and intelligent systems and are applied, for example, to autonomous vehicles and manufacturing industry.

3. Sports Management: Concept and Specificities

3.1 Definition of the concept

Organizational management can be defined as: "The overall steering of the organization through a set of policies for the production of goods or services, communication, marketing, human resources, financing policy, budgetary control [...] that are coherent with each other and converge towards the strategic project and are reflected in the organizational culture." (Bayle, 2007, p. 60).

Sports management is “the art and science of directing and managing non-profit organizations that organize sports activities in such a way as to maintain their general interest objectives, while accepting their economic imperatives.” (HAMDOUS, 2018, p. 5).

Sports management is the set of practices aimed at leading, directing, structuring and developing sports organizations of all kinds.

GARY TRIBOU, for his part, believes that "offering sports services or even managing a sports service is not simply a matter of applying management tools. For him, there is a specificity in the demand for sport that comes from users, public services, or club members, as well as from the clients of commercial service providers who call for a specificity in the offer and its management tools. (Tribou, Dermit-Richard, & Wojak, 2015, p. 3)"

In light of all these definitions, two key aspects of sports management emerge (HAMDOUS, 2018, p. 6):

- On the economic level: efficiency through achieving objectives at the lowest cost and optimizing allocated budgets.
- On the social level: maintaining socio-cultural cohesion, social harmony, and the motivation of all staff.

Thus, the authors define sports management as follows: "Sports management" is "the set of specific means, processes, and techniques used within a sports organization, mobilizing all available resources with the aim of maximizing results while minimizing costs […] Rationalization of practices, both within and outside of sports practice."

3.2 The Specificities of Sports Management

Today, sport is an economic sector in its own right and deserves to be a subject of research as such. While it presents a clear multidisciplinary interest—economic, but also philosophical, sociological, anthropological, psychological, legal, and political—and even interdisciplinary, it requires reflection on its management methods, commensurate with its importance in the markets (labor market, product market, financial markets, etc.). (Lardinoit & Tribou, 2004, pp. 127-128):

Sport is not an ordinary industry. The very existence of a sporting product (a competition) requires at least two competitors. But while a monopolistic situation is antithetical to sporting competition, it is nevertheless essential at the level of market-regulating institutions (FIFA, IOC, etc.) in order to impose sporting rules that guarantee the sustainability of competitions.

Another specificity: competition is confined within categories such as the first and second football leagues in Europe or the major and minor leagues in the United States, and does not extend from one category to another. Whereas in the business world, an SME can very well compete with a multinational in certain markets.

Production is also specific. A sporting spectacle is intangible, ephemeral, unpredictable, and subjective. It is simultaneously shaped and consumed. The interactions between spectators-consumers and athletes-producers are constitutive of the good and contribute as much to the perception of the quality of the show as the competition itself. The implications for the marketing approach are therefore complex.

In terms of communication, sponsorship also presents specific characteristics. Indeed, sponsorship persuasion processes rely on the associations established by consumers between the sponsored sport and the sponsoring brand. The psychological processes of associations are highly complex. Ignoring them, on the one hand, significantly compromises the effectiveness of this communication technique, and on the other, leads to the use of measurement tools whose validity is largely unsatisfactory.

There is another area where sports management presents a remarkable specificity: that of the associative and semi-professional offering of sports services. 2 Indeed, managing the human resources of a sports club or federation is not like managing a trivial business. The presence of volunteers, permanent employees, and elected officials makes human capital management particularly delicate.

In the realm of ethics, sport and its use by businesses also present specific characteristics. While spectator sport, with its media and economic corollary, gives sport most of its social visibility today, the educational, medical, and social functions of amateur sport are still active in our societies. Professional sport cannot ignore the educational function of competition, just as physical activities for social integration cannot be organized on the margins of economic reality. Sport is "embedded" in the modern socioeconomic fabric. This interdependence, too quickly mentioned here, suggests that not having access to a conceptualization of the different functions of sport and their interrelationships in socioeconomic life can generate ethical tensions that are detrimental to stakeholders in the sports economy.

3.3 Common characteristics of sports organizations

Sports organizations are distinguished by four differentiating elements (Bayle, 2007, p. 66):

- A purpose other than the priority or systematic pursuit of profit (extra-economic or even societal in nature);
- Financing based on a mixed economy (presence of direct and indirect public funding, as well as more or less significant funding from the "commercial" sector);
- A mixed status of the personnel driving their management (coexistence of paid staff, volunteers, and sometimes staff provided by the state and/or local authorities or even public companies);
- Membership in national but also supranational regulatory systems (international federations, International Olympic Committee) claiming autonomy from the public sphere.

Table 1. The four levels of classification of organizations in the field of sports management

Level 01	**Organizations at the heart of the sports sector (sports movement organizations)** Sports federations and associations affiliated with federations Professional sports leagues and clubs
Level 02	**Other sports sector organizations** Sports service companies (sometimes approved by a federation) Sports event organizers (sometimes registered on federal calendars) Sports associations not affiliated with a sports federation
Level 03	**Sports-related organizations** Ministry of National Education, Ministry of Youth, Sports and Community Life, etc.... Local government sports departments

Level 04	Sports consulting firms, sports media, specialized sports communications agencies, etc. Organizations **(unrelated to sports) using sports as a management tool** Large companies in particular, and more broadly, any public or private organization

Source: (Bayle, 2007, p. 66)

3.4 The Scope of Sports Management

Management systems focus on management practices within organizations and their relationship to the effectiveness and performance achieved. The term "sports organizations," used in French and Anglo-Saxon studies, in sociology, and sports management, has been used in different ways. According to Chantelat, Garrabos, & Pigeassou9, "sometimes it refers solely to organizations within the sports movement, sometimes to the broader concept of sports service organizations, or even, more generically, to any organization directly or indirectly involved in physical and sporting activities (sports associations, sporting goods manufacturing companies, etc.)." The diversity of the organizations studied in terms of size, legal status (private/public; company/association), or purpose (commercial/non-commercial) raises the question of the specific management principles and practices that should be applied to them (HAMDOUS, 2018, p. 6).

4. Opportunities offered by AI in sports management

AI is opening up vast possibilities in the field of sports, offering unprecedented opportunities to improve athlete performance, prevent injuries, and strengthen fan engagement. Thanks to technological advances, AI enables increased training personalization, optimized game strategies, and more immersive interaction between sport and its audience. (Naully, 2025)

- By leveraging machine learning models, AI can deeply analyze athletes' biometric data and movement patterns to optimize their technique and training. For example, at the Paris 2024 Olympic Games, AI was used to provide real-time recommendations to coaches and athletes, improving the accuracy and effectiveness of training programs.

- In disciplines such as swimming and athletics, AI has been integrated into body sensors to analyze movements and adjust an athlete's posture or technique. This approach not only helps improve individual performance, but also compare strategies and identify areas for improvement. Companies such as SAP and IBM are already developing AI-powered solutions to evaluate player performance in real-time in team sports such as football and basketball.
- AI also plays a key role in injury prevention by anticipating risks related to muscle fatigue, postural imbalances, or excessive training loads.
- Thanks to smart sensors and physiological data monitoring, athletes benefit from a more proactive approach to managing their recovery. These tools monitor indicators such as heart rate, heart rate variability, and lactate levels, providing a detailed view of an athlete's physical condition. This predictive analysis significantly reduces athletes' downtime and helps extend their careers.
- AI is also transforming the spectator experience by offering interactive and immersive ways to follow competitions. Fan engagement is enhanced through advanced analytics and enriched digital experiences.
- AI-powered augmented and virtual reality technologies allow fans to relive key moments from different perspectives, while intelligent chatbots provide personalized information about ongoing competitions. Platforms such as Google DeepMind's have been used to offer live predictions on match results and event trends, improving the experience for spectators around the world.
- AI is enabling greater accessibility to sports competitions by automatically translating commentary and generating real-time subtitles for international events, making sport more inclusive and accessible to a global audience.

5. Current challenges of AI in sports management

AI in sports management is not without controversy. Its integration raises several major challenges that must be addressed to ensure its fair, ethical, and effective use (Naully, 2025):

- One of the most pressing challenges concerns the fairness and integrity of competitions. AI provides athletes and teams with advanced tools for analyzing performance and optimizing strategies, but its unequal access can lead to competitive imbalances. According to the International Olympic Committee (IOC, 2024), the Olympic AI Agenda establishes guidelines to ensure that AI does not confer unfair advantages on certain nations or athletes with greater financial resources ;
- Deloitte AI Institute emphasizes the importance of a clear regulatory framework, similar to anti-doping regulations, to prevent the misuse of AI for performance enhancement;
- The massive use of athletes' biometric and behavioral data poses a major ethical and legal challenge. Data collection via body sensors and connected equipment must respect the principles of confidentiality and informed consent. The 2024 Olympic Games have seen the introduction of strict data protection protocols, ensuring that this information is only accessible to authorized technical teams and is anonymized to prevent misuse;
- The US AI Institute highlights the risk of cyberattacks and sensitive data leaks that could compromise not only the privacy of athletes but also the integrity of competitions. It is therefore essential that sports organizations collaborate with cybersecurity experts to minimize these risks;
- Another fundamental challenge concerns the authenticity of sport. By improving the accuracy of refereeing decisions and optimizing athletes' performances, AI could transform competitions into a clash between technologies rather than between individuals.
- The IOC insists that these technologies must remain assistive tools and not completely replace human judges. Strict regulation and

ongoing reflection are necessary to preserve the spontaneity and emotion inherent in sport.

6. CONCLUSION

The integration of AI into sports management marks a major transformation that redefines the boundaries of human performance, spectator experience, and refereeing decisions. As technology continues to evolve, the use of AI and data in sports management will likely become even more crucial. Teams that adopt these tools will be better positioned to meet the challenges of modern sport and maximize their chances of success.

However, its use raises ethical and practical debates that require careful consideration of the future of sport and the values it embodies. It is imperative that this evolution be guided by clear ethical principles and continuous reflection on the boundaries that must not be crossed.

Sport has always been a field of innovation, but it must remain, above all, a space where human skills, resilience, and effort are valued. AI must be a support tool and not a substitute for human decision-making. Therefore, sports federations, athletes, and regulatory bodies must collaborate to integrate AI in a thoughtful and balanced manner, ensuring a future where technology enhances sport without altering its essence.

Ultimately, the sports management of tomorrow will not only be a physical or tactical confrontation, but also an interaction between human and artificial intelligence. The challenge is to find the right balance where AI complements and enriches sport, without distorting it. The path is clear, but it will depend on the choices we make today to preserve the fundamental values that make sport what it is: a challenge where humans remain at the center of the game.

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